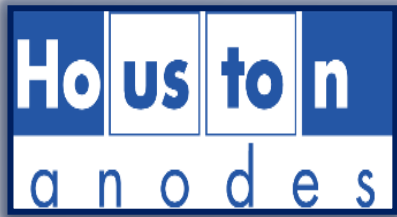




Advantages of Manufacturing Sacrificial Anodes for Windfarms with Induction Melting Technology





OVERVIEW

Company Summary

Safety

Corrosion

Cathodic Protection

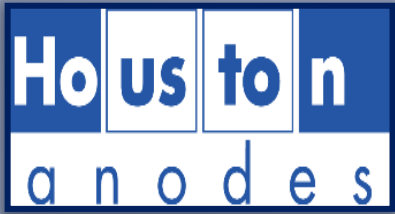
Quality Control

Raw Materials

Induction Melting Technology

Our Deliverables





HOUSTON ANODES GROUP

Protecting Offshore, Subsea and Onshore Assets and Infrastructure since 1977.

Designing, Manufacturing and Delivering High-Quality Aluminum and Zinc Sacrificial Anodes.

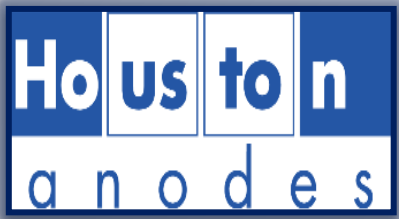
Equipped with a state-of-the-art laboratory and superior induction melting technology.

Headquartered in the Energy Corridor of Houston, TX.

Management System Certificate ISO 9001-2015 by DNV.

CLOSE TO OUR CUSTOMERS





HSE POLICY STATEMENT

At Houston Anodes International, LLC. we are proud of our holistic approach to **safety in the workplace**. To that end, we commit to ensure **no harm** comes to **our employees**, or damage to the **environment**.

We will endeavor to **prevent all labor-related accidents** and to continuously improve our HSE performance through the **active participation** and support of all our stakeholders.

The **successful management of HSE** is an essential part of our business model and **corporate culture**.



HEALTH, SAFETY AND ENVIRONMENT (HSE) POLICY STATEMENT

Houston Anodes International, LLC is committed to protecting the health and safety of all employees and preventing accidents and loss of any of our assets.

Management is committed to providing a safe work environment that is in keeping with industry standards and all federal and state regulations. The company will always strive to eliminate any and all hazards that could endanger our employees or damage personal or company property in any way.

All employees have a responsibility to always do their best to reduce the possibility of accidents or injury in the workplace. Work practices and safety procedures are outlined in our company Employee Handbook.

Through a collective effort between the company and its employees, the possibility of accidental loss can be minimized and /or eliminated. Safety is everyone's responsibility, not only that of supervisors and managers.

All employees are required to comply with all safety rules and requirements with regard to the operation and maintenance of company owned assets. The employees have the responsibility to perform their jobs according to all safety rules and regulations. To meet our goals and objectives, we will:

1. Comply with applicable HSE legislation and regulations and apply responsible standards where these do not exist.
2. Identify HSE risks arising from our activities and reduce them to the lowest practicable levels.
3. Ensure visible and supportive management involvement in all HSE matters.
4. Provide a safe working environment for all our employees and ensure personnel are trained to carry out their work safely.


Chief Executive Officer
Houston Anodes International, LLC

DEC 1, 2015
Date

REV. 8

REV. DATE: DEC, 2015

SAFETY IS OUR # 1 PRIORITY

CORROSION

Metals extracted from its primary ore (**metal oxides**) have a **natural tendency to revert to that state** under the **action** of **oxygen** and **water**.

Corrosion is an **electro-chemical process** that involves the passage of electrical currents on a micro or macro scale by an “**anodic**” **reaction**.

This reaction **produces free electrons**, which pass within the metal to another site on the metal surface (**the cathode**), where it is consumed by the “**cathodic**” **reaction**.

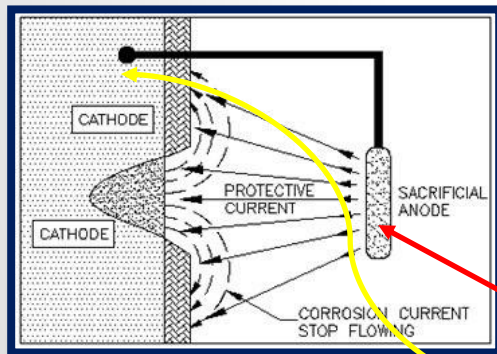
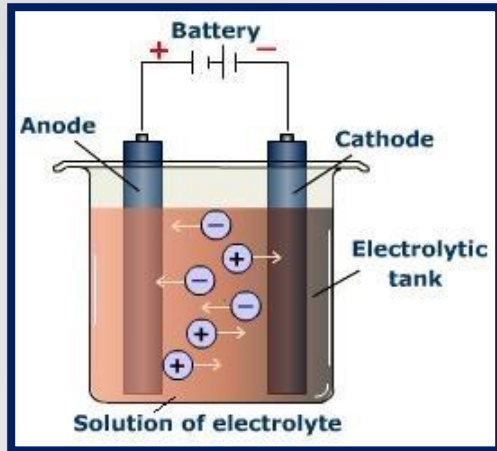
Corrosion process is initially caused by:
Difference in **natural potential** in galvanic (bimetallic) couples. **Metallurgical variations** in the state of the metal at different points on the **surface**.

Oxygen **rich** areas become the **cathode** and oxygen **depleted** areas become the **anode**.

CORROSION NEVER SLEEPS



How to anodes



CATHODIC PROTECTION

The principle of **Cathodic Protection** is in **connecting** an external **anode** to the **metal to be protected** and the passing of an **electrical dc current** so that all areas of the metal surface become **cathodic** and therefore do not corrode.

The **current** is a result of the **potential difference** between the two metals.

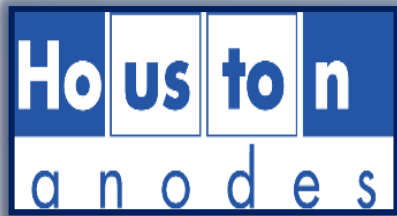
Causes a **positive current** to flow in the electrolyte, from the **anode** to the **steel**.

Surface of the **steel** becomes **more negatively charged** and becomes the **cathode**.

A **Galvanic System** requires:

- i) **Sacrificial anodes**
- ii) **Direct welding** to the **structure** or a **conductor** connecting the **anode** to the **structure**

SIMPLE AND EFFECTIVE



HOUSTON ANODES GUARANTEE

Homogeneous alloys always conforming to **exact international quality standards and specifications.**

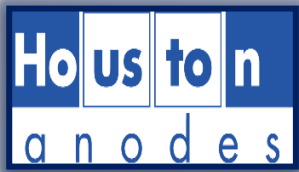
**NACE SP-0387 & SP-0492, DNV RP B401, ISO 155589-2,
NORSOK M-503**

Houston anodes are fabricated using only **raw materials** of the **highest purity**, assuring **superior electrochemical and chemical properties.**

Our team of professional engineers, manufacturing and sales personnel are committed to provide the best products with **unparalleled customer service.**

NOT A SINGLE CLAIM IN 40 YEARS





QUALITY CONTROL

In-house state-of-the-art Laboratory equipped with a Bruker-Quantron Magellan Q8 spectrometer.

All raw materials inspected and tested upon arrival **prior to** release for production.

Chemical results verified against MTR's.

Chemical analysis performed on every heat.

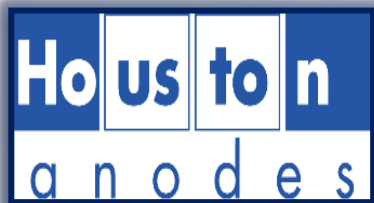
Electrochemical properties performed on every batch according to **NACE TM-0190.**

Rigorous Inspection and Testing Points.

Scheduled equipment calibration per **OEM.**



**IMMEDIATE ACCURATE
TEST RESULTS**



RAW MATERIALS

Primary High-Grade Raw Materials

Premium US and global suppliers

Primary Aluminum - 99.9% purity

Special High Grade Zinc - 99.99% purity

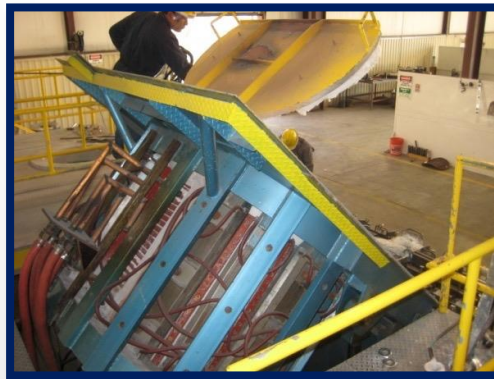
Indium - 100 % purity

ASTM Steel - anode cores & mounting structures

**NEVER MELT SCRAP OR
RECYCLED ANODIC MATERIAL**



ELECTROMAGNETIC INDUCTION TECHNOLOGY



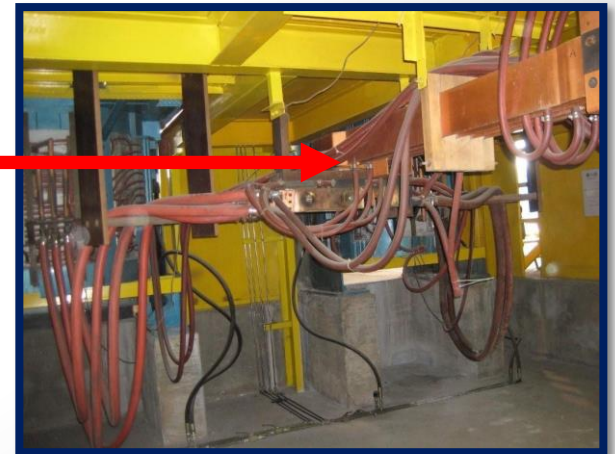
HOW IT WORKS

The furnace is composed of a **deionized water-cooled copper coil**.

When energized with an AC power source, it develops an **electromagnetic field**.

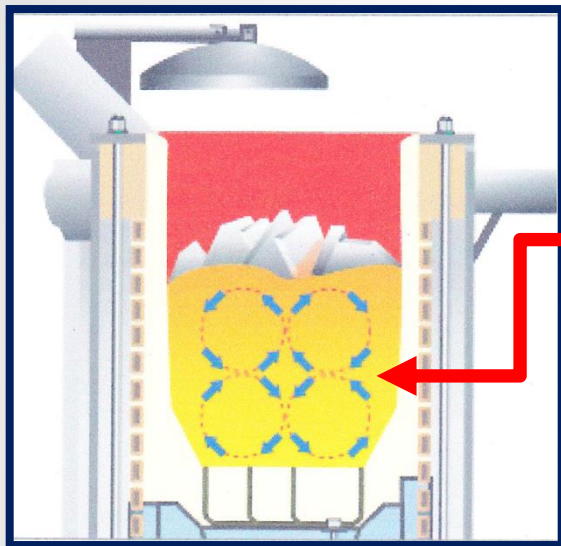
Any metal charge kept inside the crucible will be subjected to electromagnetic induction.

The electromagnetic field creates **EDDY currents** in the charge, that causes an **opposite circulating stirring flow** of electrons within the molten mass, which in turn, generates a **huge amount of heat** in the charge depending on its resistivity.



**HEAT GENERATED THROUGH
ELECTROMAGNETIC INDUCTION**

INDUCTIVE STIRRING CIRCULAR EDDY CURRENTS



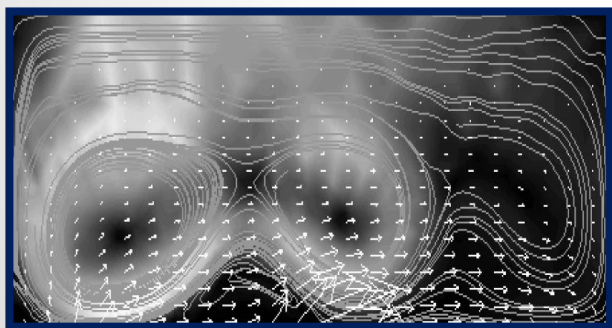
The electrical field circulating through the molten metal creates a **stirring action** that produces a higher-quality alloy.

Once the metal is molten, the **electro-magnetic field** causes the bath to move in a “**Figure 8**” pattern.

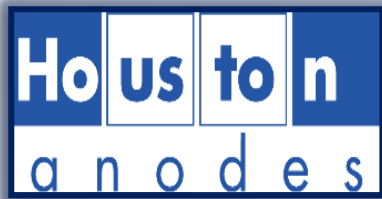
Inductive stirring is the **ONLY** process capable of incorporating **INDIUM 100%** homogeneously at a molecular level

The amount of stirring is determined by the size of furnace, power, frequency of the electromagnetic field and type of metal.

Inductive stirring **circulates** the molten metal **away** from the furnace walls, **preventing overheating**.



NO CONTACT STIRRING



ADVANTAGES

Guarantees an 100% homogeneous alloy of exceptional purity of the electrochemical composition

Controlled temperature melting process.

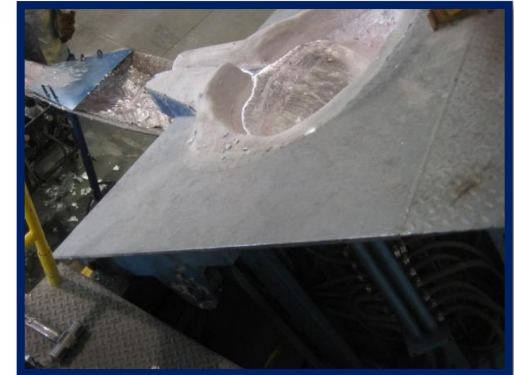
The process assures **identical results**.

Consistent **batches** and final product **every time**.

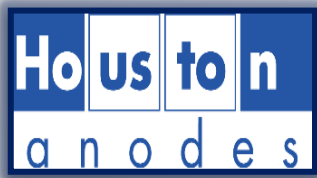
Heat is produced **instantly** and within the metal being melted.

Without flames and without touching the material to be heated.

FAST CYCLE TIMES



**RELIABLE, REPEATABLE, EXTREMELY EFFICIENT
NON-CONTACT MELTING IN MINIMAL AMOUNT OF TIME.**



MORE ADVANTAGES

**FREE OF HYDROGEN & CARBON BUBBLES
POROSITY**

Favorable working conditions

Doesn't heat up surrounding area or environment.

ENVIRONMENTALLY FRIENDLY

Clean and odorless heating process.

No harmful emissions or noise

No polluting fuel or burners

ENERGY EFFICIENT

Power is on only when being utilized.

REDUCED METAL LOSS.

Better product yield.



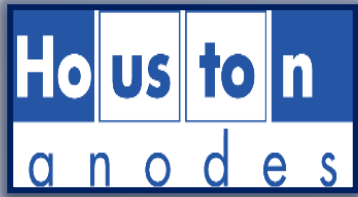
SAFER, EMISSIONS FREE, REDUCE METAL LOSS

INDUCTIVE STIRRING (EDDY CURRENT)

**ONLY PROCESS
CAPABLE
OF INCORPORATING
INDIUM
100%
HOMOGENEOUSLY
AT A MOLECULAR
LEVEL**



(CLICK VIDEO TO START)



HOUSTON ANODES ELECTROMAGNETIC INDUCTION FURNACES

TOTAL CAPACITY: 625 TONS/MONTH
INDUCTION FURNACE 1: 400 KW MELTING
CAPACITY: 250 TONS/MONTH
INDUCTION FURNACE 2: 625 KW MELTING
CAPACITY: 375 TONS/MONTH

Furnace operates at a range of 0-2400v, 0-625kw, and 0-2000Hz .

Controlled temperatures vary depending on the type of anode.

Eliminates the risk of superheating the metals



**PRECISE MELTING
CONTROL**

ENHANCED SAFETY

GAS COMBUSTION FURNACES

Heat is created by applying **direct flame** to a crucible and the aluminium blocks, once the crucible is hot, it transfers the heat to its interior where the alloys are placed to be melted by convection

Gas Furnace flame directly transfers Hydrogen & Carbon to the metals when melting, **contaminating** the chemical alloy adding impurities, and creating bubbles that became **porosities** when the alloy is cooling.

In a Gas Furnace **the mixture is made by Hand Stirring** with a steel rod (Extremely Dangerous) which leaves particles of Iron, **poisoning** the chemical alloy

Gas Furnace produces **only Heterogeneous mixture of chemical alloy**, where the chemical activator **INDIUM** is just **partially** induced, creating high percentages of passive or **inert** portions in the alloy, reducing the lifespan of the anodes and considerably **affecting the cathodic Protection of the Asset.**



INDUCTION VS GAS ADVANTAGES

NON-INSTALLED ANODE

**ANODE MELTED BY GAS
FURNACE**

COMPLETE LIFETIME ANODE

59% WORN-OUT MATERIAL

41% NON USEFUL MATERIAL

COMPLETE LIFETIME ANODE

90% WORN-OUT MATERIAL

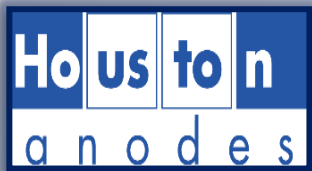
**ANODE MELTED BY
ELECTROMAGNETIC
INDUCTION TECHNOLOGY**

COMPLETE LIFETIME ANODE

100% WORN-OUT MATERIAL

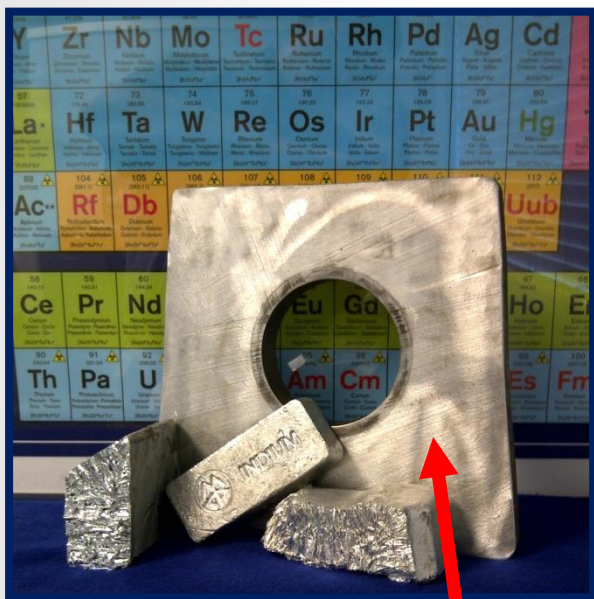
0% NON-USEFUL MATERIAL





HOUSTON ANODES DELIVERS

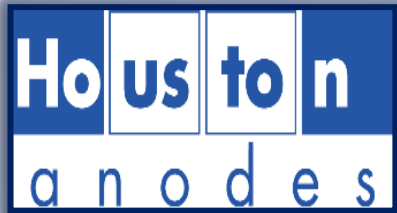
RIGHT CHEMICAL & ELECTROCHEMICAL PROPERTIES



**NO POROSITY,
PERFECT DENSITY**



**COMPLIANT WITH:
NACE SP 0387; SP 0492
DNV RP B401
ISO 15589-2:2012
NORSOK M-503
AND ANY OTHER
CUSTOMER'S SPECIFICATIONS**



HOUSTON ANODES DELIVERS

RIGHT WEIGHT



EXACT ANODIC MASS

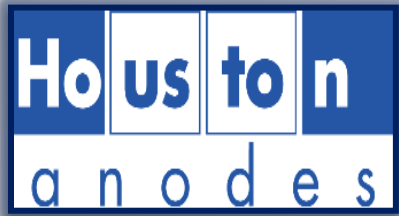
HOUSTON ANODES DELIVERS

RIGHT FIT



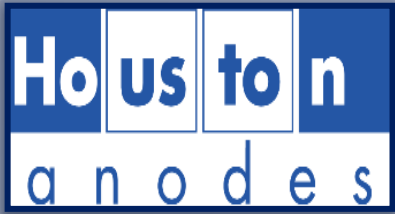
**EASY FIT
FOR INSTALLATION**





OFFSHORE WINDFARM SACRIFICIAL ANODES





MANUFACTURED ANODES FOR OFFSHORE WINDFARM PROJECTS:



**RHODE ISLAND:
REVOLUTION WIND**

**NEW JERSEY:
OCEAN WIND 1**



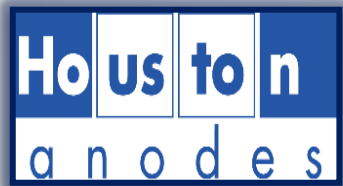
GLOBAL PRESENCE

Successfully delivered Sacrificial Anodes worldwide to Top 100 Energy Companies



APPROVED FMC GLOBAL SUPPLIER

**SUPPLIER OF 80% OF THE SACRIFICIAL
ANODES INSTALLED ON PEMEX
PROJECTS OVER 35 YEARS**



SUCCESS STORIES

Houston Anodes currently **protecting major Offshore Assets** and **subsea structures** throughout the world for valued clients including among others:



OFFSHORE AND SUBSEA PROJECTS



CERTIFICATES



Houston Anodes
PRELIMINARY QUALITY CERTIFICATE NO. GAUS/18

ITEM	TEST METHOD	TEST RESULT	TEST METHOD	TEST RESULT	TEST METHOD	TEST RESULT	TEST METHOD	TEST RESULT
Element	Specification	Heat	Heat	Heat	Heat	Heat	Heat	Heat
Aluminum	0.12% Max	0.082	0.082	0.082	0.082	0.082	0.082	0.082
Iron	0.05% Max	0.017	0.017	0.017	0.017	0.017	0.017	0.017
Carbon	0.01% Max	0.007	0.007	0.007	0.007	0.007	0.007	0.007
Copper	0.01% Max	0.007	0.007	0.007	0.007	0.007	0.007	0.007
Chromium	0.01% Max	0.007	0.007	0.007	0.007	0.007	0.007	0.007
Others	0.01% Max	0.007	0.007	0.007	0.007	0.007	0.007	0.007

Houston Anodes
QUALITY MANUAL

REV.	DATE	STATUS	APPROVED BY
01	09-18-2011	Approved for Internal Use	Antonio Galvan, President
02	09-18-2011	Approved for Internal Use	Antonio Galvan, President
03	02-06-2012	Approved for Internal Use	Antonio Galvan, President

Cunningham Business Park
 6425 Cunningham Rd, Houston, TX 77041
 PHONE (832) 245-9701 E-MAIL: gaus@gausanodes.com WEBSITE: www.gausanodes.com



ORSTED APPROVED:

D-QA-GR-016 Audit report subcontractor / supplier - Summary

COORDINATES

Name subcontractor: Houston Anodes Contact person: Antonio Galvan (CEO), Jaime Galvan (Business development)
Audit date: 21/01/2022 Auditees: Jaime Galvan (Business development), Sully de Jesus (QA/QC)
Auditor: Tim Balcaen Project: 25280 Southfork & Revolution
Address: 6425 Cunningham Road
Houston, Texas 77041 (USA)

AUDIT RESULTS

	Quality	Health & Safety	Environment	VCA SCC*	minimum target	desired target
1. QHSE management system	91%	89%	85%		50%	80%
2. Management Responsibility	98%	85%	85%		50%	80%
3. Resource management	100%	86%	88%		50%	80%
4. Product realisation	100%	100%	94%		50%	80%
5. Monitoring, control and continuous improvement	88%	77%	75%		50%	80%
OVERALL SCORE	95%	84%	81%	76%	50%	80%

VCA annex A "must for in-house subcontractors"

1. QHSE management system

2. Management Responsibility

3. Resource management

4. Product realisation

5. Monitoring, control and continuous improvement

6. Health & Safety

7. Environment

8. minimum target

9. desired target

10. Quality

11. Health & Safety

12. Environment

13. minimum target

14. desired target

15. Quality

16. Health & Safety

17. Environment

18. minimum target

19. desired target

20. Quality

21. Health & Safety

22. Environment

23. minimum target

24. desired target

25. Quality

26. Health & Safety

27. Environment

28. minimum target

29. desired target

30. Quality

31. Health & Safety

32. Environment

33. minimum target

34. desired target

SUMMARY

Quality (both QA and QC) at a very high level. Follow-up seemed very strict and all things under control.

Most comments are regarding HSE. There is a lot of HSE knowledge due to the great expertise and the small scope (anode production). However, some of the HSE foundation is missing: risk analysis on multiple layers, HSE as integral part of company processes (process overview, organization chart, management review).

Some of the remarks stated in the report might be overkill for such a small organization. Interpret at own discretion :-).

Some NCs on the next tab are left without an action defined. This means the audit question is not positively answered, but no immediate action is required.

D-QA-GR-016

Audit report subcontractor / supplier - Summary

ACTION

NC = non-conformity, OFI = opportunity for improvement, BP = best practice

NC	add HSE as a staff function to the organization chart
NC	improve on HSE risk analysis * perform company level HSE risk analysis, e.g. fire * draft JHAs * add categories of vulnerable personnel as a topic to analyse in the JHAs
NC	define other waste streams than scrap, at least single-stream recyclables
NC	define a method for root cause analysis
OFI	* add HSE as a staff process to the process overview * post KPI statistics on the workforce * add more HSE topics to the management review * add confined space blasting as a standard JHA / JSA * have a central overview of ongoing actions instead of (on top of) actions in all the separate documents
remarks from walkabout	* securing of gas bottles in storage * welder qualification / welding equipment calibration not checked during audit due to limited time and not part of project scope
Remark	Be aware of the very strict specs towards as-built documentation
Remark	Make sure that the absence of medical checks is in accordance with OSHA regulations. I find it very unusual, so I have marked it as an NC in the report.

APPROVAL

Smulders Group
Name:
Signature:

Contractor/ Supplier
Name: ANTONIO GALVAN
Signature:

How to anodes



THANK YOU!

